

B.Tech. Degree VII Semester Special Supplementary Examination

April 2016

IT 1705 E2 CRYPTOGRAPHY AND DATA SECURITY

(2012 scheme)

Time : 3 Hours

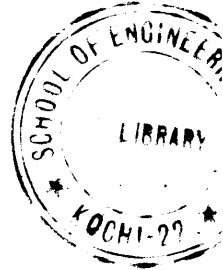
Maximum Marks : 100

PART A

(Answer ***ALL*** questions)

(8 × 5 = 40)

- I. (a) Explain cryptography and cryptanalysis.
- (b) Write short notes on error probability and security.
- (c) Differentiate stream and block enciphering.
- (d) Define non-linear shift registers.
- (e) Write short notes on knapsack system.
- (f) Define zero knowledge techniques.
- (g) Explain the general aspect of key management.
- (h) Write short notes on fair crypto systems.



PART B

(4 × 15 = 60)

- II. (a) Give the concept of Cipher system. What is (i) Transposition cipher (8)
(ii) Substitution cipher?
- (b) Explain information measure and absolute security. (7)
- OR**
- III. Explain the working of Hagelin machine with neat diagram. (15)
- IV. Illustrate the working of DES algorithm with a neat sketch. (15)
- OR**
- V. Explain random properties of shift register sequences. (15)
- VI. Explain public key system with the help of RSA algorithm. Mention its merits and demerits. (15)
- OR**
- VII. (a) Explain different cryptographic protocols. (10)
(b) Explain principles of digital signatures. (5)
- VIII. Describe the different methods used for key distribution in symmetrical and asymmetrical systems with necessary examples. (15)
- OR**
- IX. What do you mean by network security? What are the ways of employing network security? (15)